

libutils: quality audit

Date: October 4, 2026 · **Sources:** `include/` , `src/` , `tests/` , `cmake/` , `CMakeLists.txt` , `.gitignore` , `README.md` , `CHANGELOG.md` , `THIRD_PARTY_NOTICES.md` , git history (196 commits)

How to read this document

1. Summary: score per category.

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Whole project, compiler warnings pass on. The metrics are heuristic, every category was sampled by hand and the false positives are not counted. Bugs and undefined behaviour are covered by the bugs audit.

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1. Summary

Category	Findings	Severity	Score / 10
Function length	6 functions over 100 lines	medium	7
Root leftovers	<code>a.out</code> , <code>tmp.cpp</code> , <code>main/client/server.cpp</code> , <code>latex/</code> (ignored by git)	low	8
Tests coverage (public names)	27 not referenced (79 %), mostly migration aliases, interfaces and vendored internals	low	8
Compiler warnings	14, all in the vendored <code>c2dmp-hsm</code>	low	9
Commits	4 / 196 with <code>type(scope)~:</code>	low	9
Licenses / docs	0	—	10
Tabs	0	—	10
<code>.gitignore</code>	0	—	10
Include guards	0 (the 2 umbrella headers are guard-less by design: they can be included again with other <code>_Section</code> macros)	—	10
Trailing spaces / <code>auto</code> / CMake sources	false positives only (banner line 3, <code>decltype</code> tests, ignored scratch files)	—	10
Headers / naming / layout	0	—	10

Global score: 9.2 / 10.

Verdict: very consistent code base (Xartania banner, class layout, `this->` , include comments, attribute macros everywhere; no `using namespace` , C cast, `NULL` or `TODO`). The library sources build without any warning under `-Wconversion -Wsign-conversion -Wshorten-64-to-32` , the license scope is explicit. Left: readability of a few long functions, the root scratch files and the vendored code.

2. Method

Item	Value
Scope	whole project, 177 files, 27 680 lines
Working tree	commit 2b46e19 + the working tree changes
Metrics	audit-quality/scripts/collect.py --build (clang++ 22, project flags)
Manual review	cpp-style , cpp-comments , cpp-class , cmake-style , git-conventions , readme-style , license on a sample of every module
Limits	function length by brace matching (approximate); <code>untested</code> is name-based, not line coverage

3. Findings

3.1 Function length (medium)

File:line	Function	Lines
src/utils/arguments/ArgParser.cpp:100	ArgParser::parseFlags_	~169
src/utils/cli/Cli.cpp:284	Cli::getInput_	~167
src/utils/manip/smanip/parser/EETPParser.cpp:104	EETP parsing	~149
src/utils/encapsulation/SharedMemory.cpp:107	SharedMemory::read_	~137
src/utils/cli/Cli.cpp:479	Cli loop	~131
src/utils/arguments/Hooks.cpp:46	default help hook	~119

Fix: split by phase (`parseFlags_` : long / short / value parsing; `getInput_` : escape sequences, history, buffer edition).
Readability only.

3.2 Root leftovers (low)

`a.out` , `tmp.cpp` , `main.cpp` , `client.cpp` , `server.cpp` , `latex/` (stale Doxygen output) are ignored by git but stay at the root. Fix: move the scratch programs into an ignored `sandbox/` folder, delete the generated `a.out` / `latex/` .

3.3 Public names not referenced by the tests (low)

Kind	Names	Note
migration aliases (deprecated)	<code>readMouseEvent</code> , <code>readAdvancedMouseEvent</code> , <code>readCursorPosition</code> , <code>setStyle</code> , <code>resetStyle</code>	forward to the tested new names
interfaces / abstract	<code>ISocket</code> , <code>IObserver</code> , <code>AObserver</code> , <code>AException</code> , <code>AParser</code>	exercised through their implementations
internal types	<code>ParsedUsageFull</code> , <code>Usage</code> , <code>SettingsHash</code> , <code>ShmMetadata</code> , <code>Slot</code> , <code>MiddlewareType</code> , <code>Char</code> , <code>demangle</code>	used by the tested code paths, not by name
vendored internals	<code>DirectSeedSequence</code> , <code>clamp_to_byte</code> , <code>make_generator</code> , <code>normalize_</code> , <code>makeLookupTable_</code> , <code>sos_embed_optimized</code> , <code>sos_extract_optimized</code>	covered through <code>sos_embed</code> / <code>sos_extract</code> and the <code>c2dmp</code> entry points

3.4 Compiler warnings (low)

14 warnings in `include/utils/algorithms/c2dmp-hsm/algorithm/{f,}optimized.hpp` (`char` → `unsigned` , `size_t` → `float`). Vendored: to fix upstream in `TsukiNi22/c2dmp-hsm` , a local fix would be erased by the next sync.

3.5 Commits (low)

Rule (`git-conventions`): `type(scope): message . 4 / 196` commits use `type(scope)~:` (`84c0f43` , `3a4f458` , `7912afb` , `26b7465`), `84c0f43` has the typo `depreacted` . Old history: only avoid the `~` marker.

3.6 False positives (not counted)

- **Trailing spaces:** all on line 3 of the Xartania banner (the ASCII art line ends with a space in the template).
- **auto:** `tests/type/Matrix.cpp:328,331`, `tests/type/Vector.cpp:171,174,329` check the deduced type with `decltype;` `main.cpp:130` is scratch code.
- **CMake sources:** `client.cpp`, `main.cpp`, `tmp.cpp` are ignored scratch files.
- **Include guards:** `c++14.hpp` → `ATTRIBUTE_CPP14_H` (a + can't be in a macro), `fixed_string.hpp` → `FIXED_STRING_H`, vendored `C2DMP_*` prefixes, generated header, guard-less umbrellas by design.

4. Priorities

1. **Split `parseFlags_` / `getInput_`.** Gain: readability of the two most complex parsers. Effort: medium, covered by the tests.
2. **Fix the 14 `c2dmp-hsm` conversions upstream** and add an SPDX line in the sync `sed` step. Effort: small, in the upstream repositories.
3. **Clean the root** (`sandbox/` for the scratch programs, delete `a.out` / `latex/`). Effort: minutes.

5. Sources

Skills / rules: `cpp-style`, `cpp-comments`, `cpp-class`, `cmake-style`, `git-conventions`, `readme-style`, `cpp-tests`, `license`.

Tools: `audit-quality/scripts/collect.py --build`, `clang++ 22.1.8`.

Related: `audit/2026-10-04-dependencies.md`, `audit/2026-10-04-tests.md`, `audit/2026-10-04-bugs.md`.